

AI Research Assistant — RAG Fundamentals

This sample research document is designed for testing a Retrieval-Augmented Generation (RAG) pipeline. It contains enough text to demonstrate PDF extraction, chunking, embeddings, vector storage, and semantic retrieval.

1. Artificial Intelligence

Artificial intelligence is a field of computer science focused on building systems that can perform tasks that normally require human-like reasoning, perception, language understanding, or decision making. Modern AI applications commonly use machine learning models trained on large datasets. Generative AI extends these systems by producing text, images, code, audio, and other forms of content.

2. Machine Learning

Machine learning is a method in which computer systems learn patterns from data instead of relying entirely on explicitly written rules. Supervised learning uses labeled examples, unsupervised learning discovers patterns in unlabeled data, and reinforcement learning learns through interaction with an environment. The quality of training data, model architecture, evaluation process, and deployment environment all influence system performance.

3. Large Language Models

Large language models, often called LLMs, are neural network models trained on large collections of text and code. They can generate responses, summarize documents, classify information, translate languages, and assist with programming. An LLM does not automatically have access to a private company database or a user's documents. Applications therefore often combine an LLM with external retrieval systems.

4. Retrieval-Augmented Generation

Retrieval-Augmented Generation combines information retrieval with text generation. A typical RAG application first receives a user question, converts the question into a representation suitable for search, retrieves relevant document chunks, and places those chunks into the context supplied to a language model. The model can then use the retrieved information when generating its response. This approach can make answers more grounded in a specific knowledge source.

5. Document Ingestion

A RAG system begins by ingesting source documents. PDF files may contain selectable text, tables, images, or scanned pages. A PDF parser can extract selectable text, while scanned documents may require optical character recognition. Metadata such as the file name, page number, document identifier, and section title can also be preserved because metadata is useful when filtering and displaying retrieval results.

6. Text Chunking

Large documents are normally divided into smaller pieces called chunks before embedding. Chunk size should be large enough to preserve useful context but small enough to make retrieval precise. Overlap between neighboring chunks can reduce the chance that an important sentence is split between two independent chunks. Recursive character splitting is one practical strategy because it attempts larger semantic boundaries before falling back to smaller separators.

7. Embeddings

An embedding model converts text into a numerical vector. Texts with related meanings tend to have vectors that are closer together according to an appropriate similarity measure. In a RAG pipeline, document chunks are embedded during ingestion and stored in a vector database. A user query is embedded at search time, and the vector database can return chunks whose vectors are similar to the query vector.

8. Vector Databases

A vector database stores embeddings and provides similarity search. ChromaDB is one example that can be used in local development. Production systems may also use services or databases such as Qdrant, Weaviate, Milvus, Pinecone, or PostgreSQL with pgvector. Selection depends on factors such as scale, latency, filtering, availability, deployment model, security, cost, and operational requirements.

9. Semantic Search

Semantic search attempts to retrieve information based on meaning rather than exact keyword matches. For example, a query asking about 'how computers learn from examples' may retrieve a document discussing machine learning even if the exact words in the query do not appear in the document. Embedding-based retrieval is a common approach for semantic search in RAG systems.

10. Evaluation and Grounding

A useful RAG application should be evaluated rather than judged only by a few manual examples. Evaluation can examine retrieval relevance, answer correctness, faithfulness to retrieved context, latency, and cost. Developers should test both successful and difficult questions, including questions for which the knowledge base does not contain an answer. A grounded system should avoid presenting unsupported information as if it came from the retrieved documents.

End of Sample Research Document

Use this document to test the AI Research Assistant pipeline. The intended workflow is: PDF loading → text extraction → chunking → embeddings → ChromaDB → semantic retrieval → RAG generation.